AI-BASED AUTOMATED DETECTION AND CHARACTERIZATION OF BONE LESIONS AND FRACTURES ON PELVIC RADIOGRAPHS

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Learning Objectives

1. Understand how deep learning can aid in automatic bone lesion and fracture detection.
2. Review pipeline design and dataset preparation
3. Evaluate performance metrics

Contents

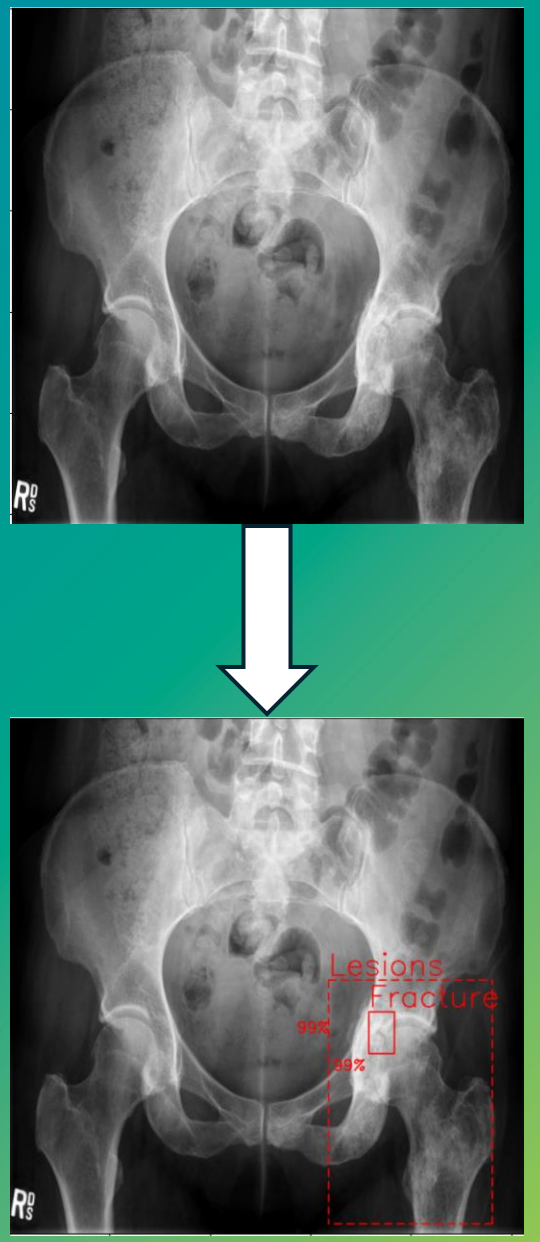
- Introduction: Bone Lesion and Fracture Detection on Hip X-ray
- Dataset description
- Methods
  - Experiments
  - Training
- Results
- Conclusion and take-home points

Introduction: Bone Lesion and Fracture Detection on Hip X-ray

Context: Rapid and accurate detection of Bone lesions and fractures is critical in clinical workflows

Problem: Time consuming, Susceptible to human Error, labor Intensive

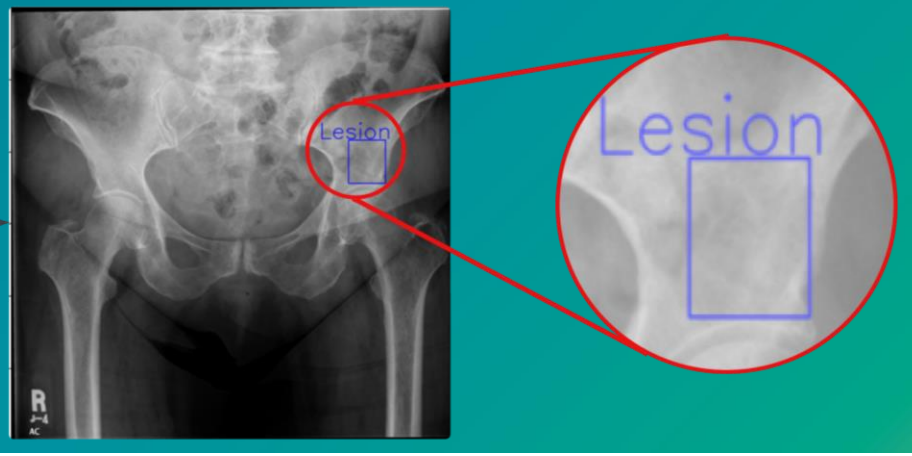
Goal: Automate Bone Lesion and Fracture Detection using AI

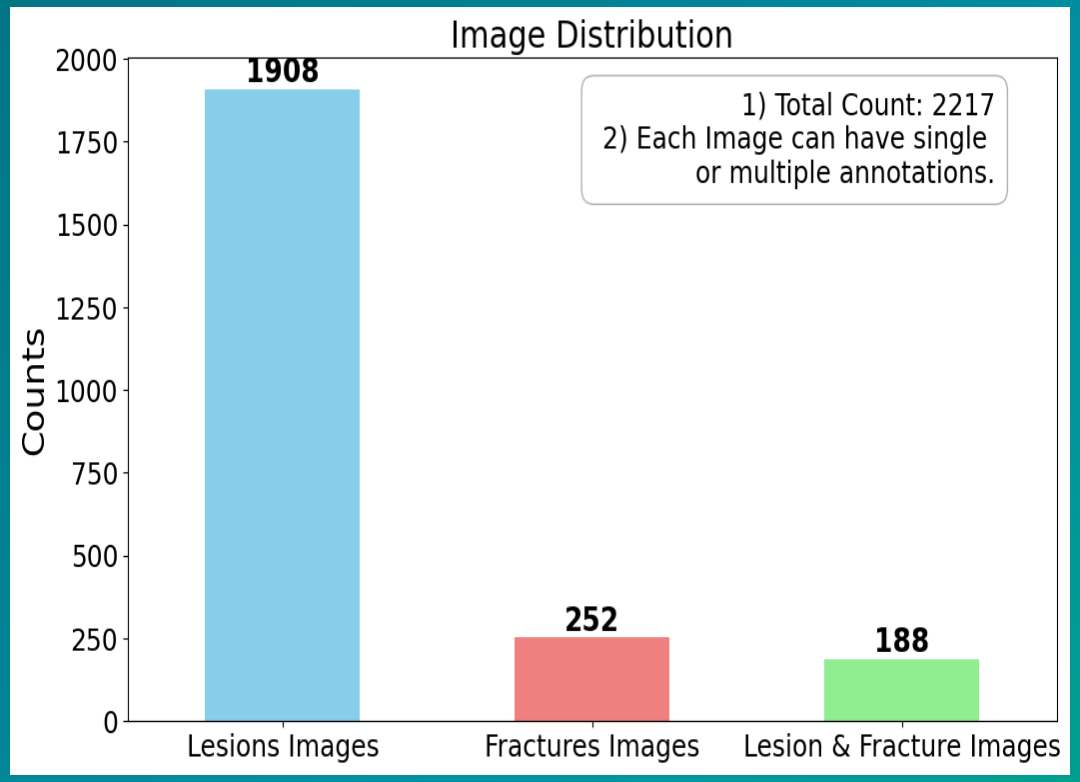


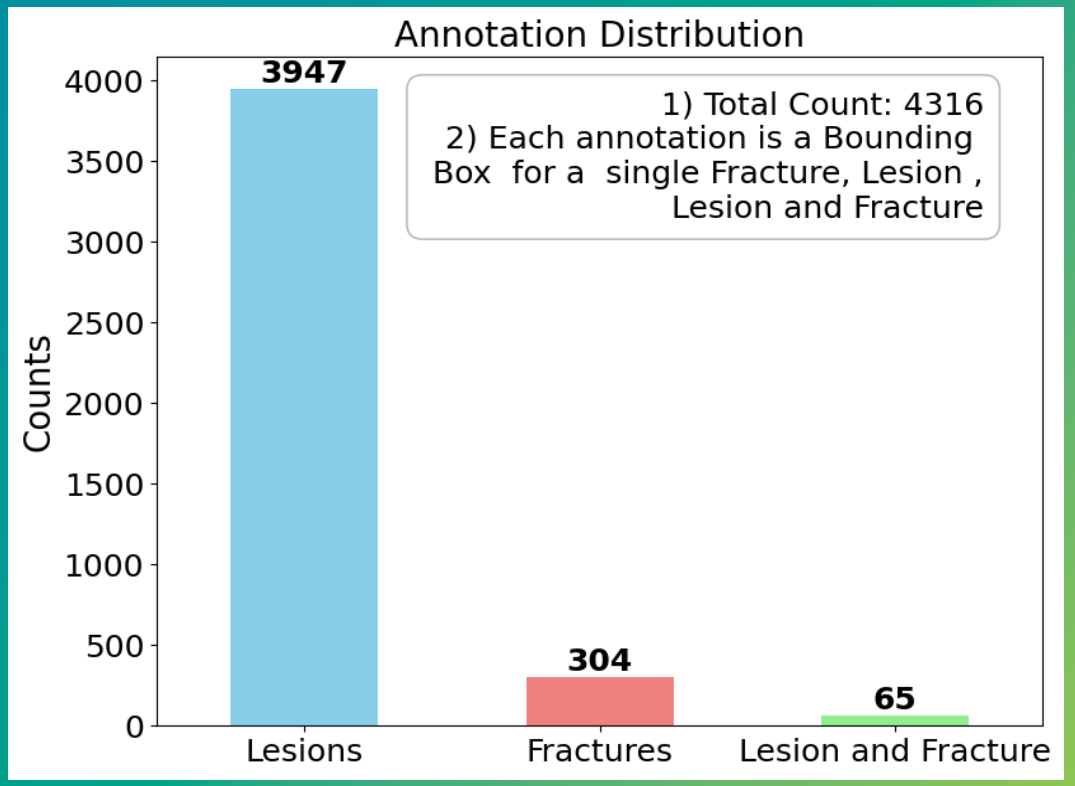
Dataset: Image/Annotation Distribution



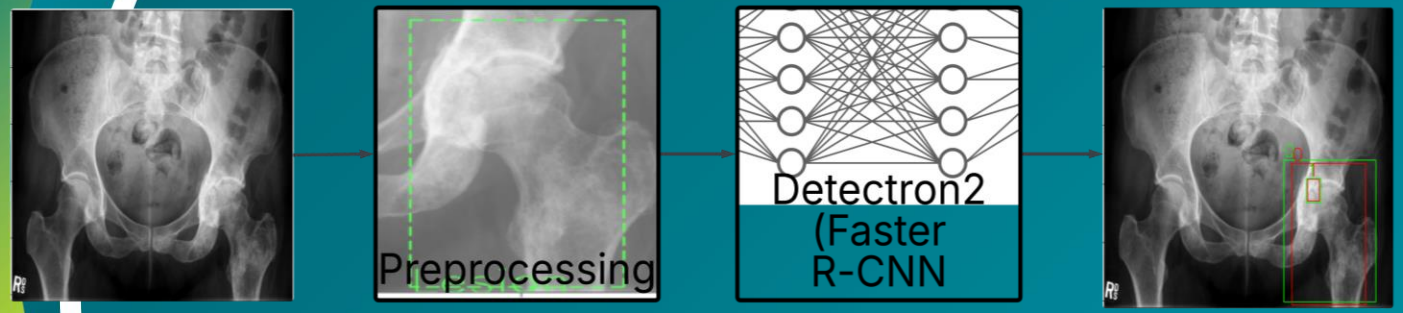




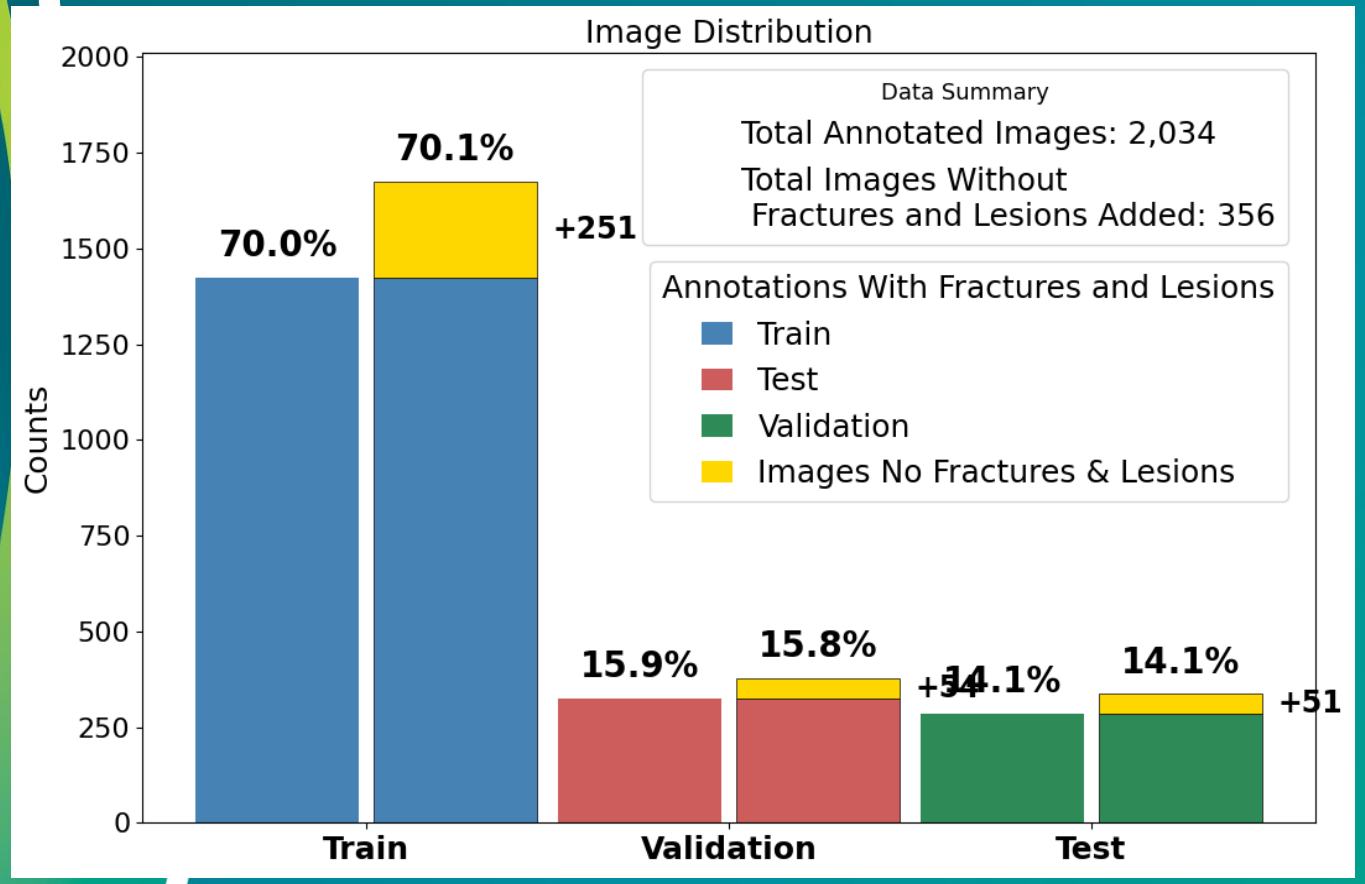




Methods/Experiments: Train/Val./Test; AI model training

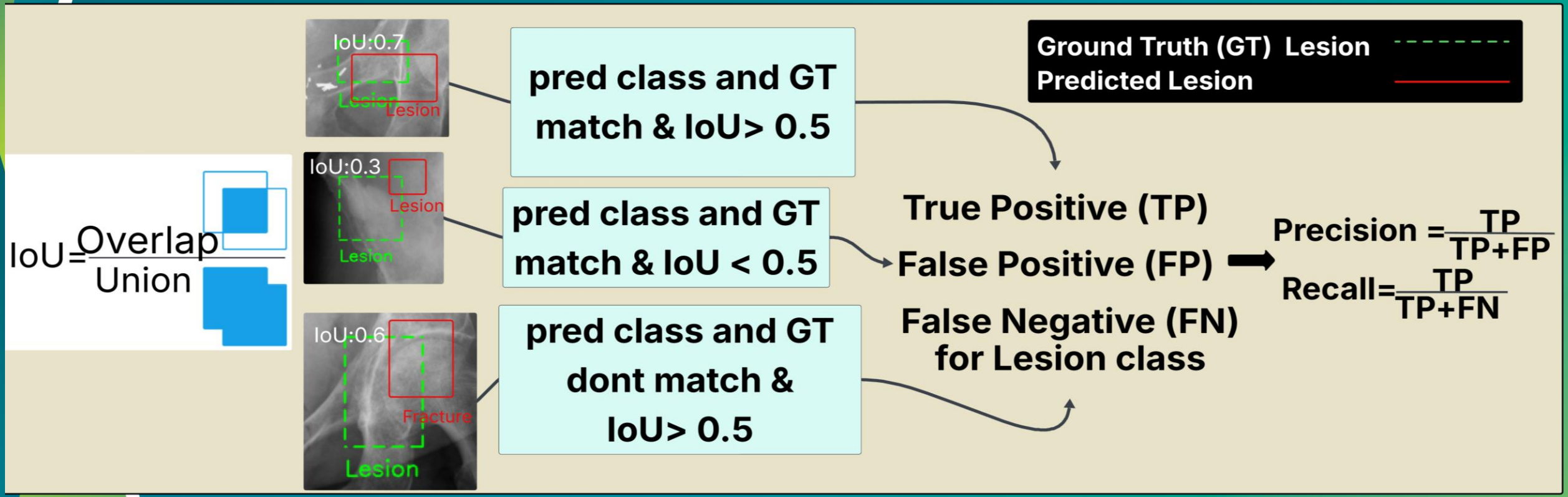


- Model:
  - Detectron2
  - Faster RCNN



- Data Split:
  - Train (70%)
  - Validation (16%)
  - Held Out Test (14%)
- Training:
  - 0% & 20% normal images
  - 350k training iteration
  - Optimal model selected based on validation set performance

Performance Eval: Precision / Recall / Average precision (AP)



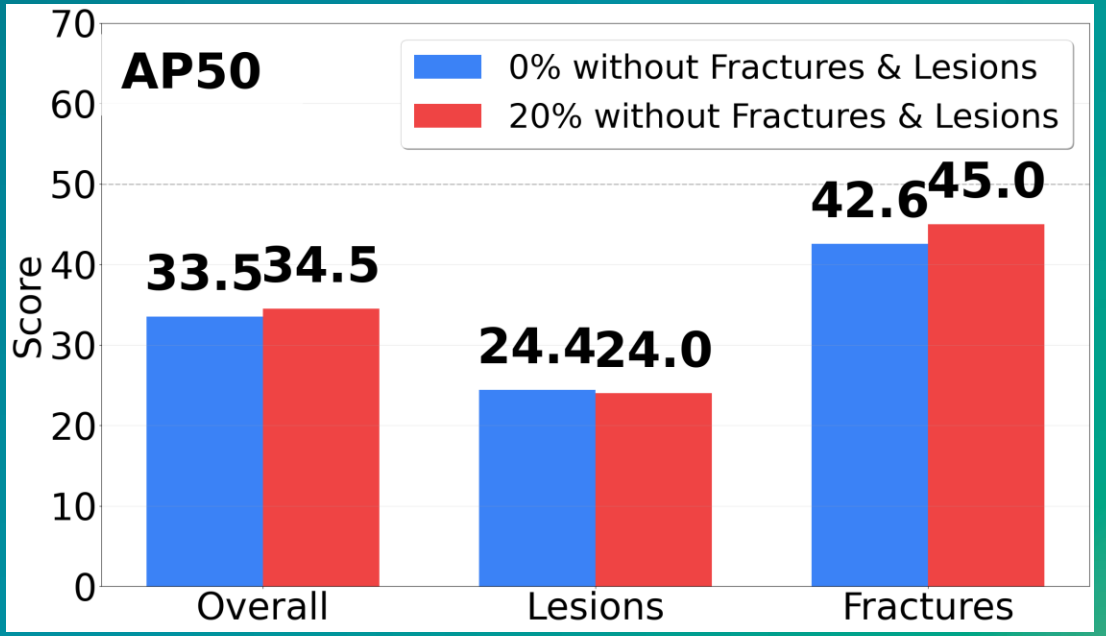
AP50: Average precision at IoU > 0.5

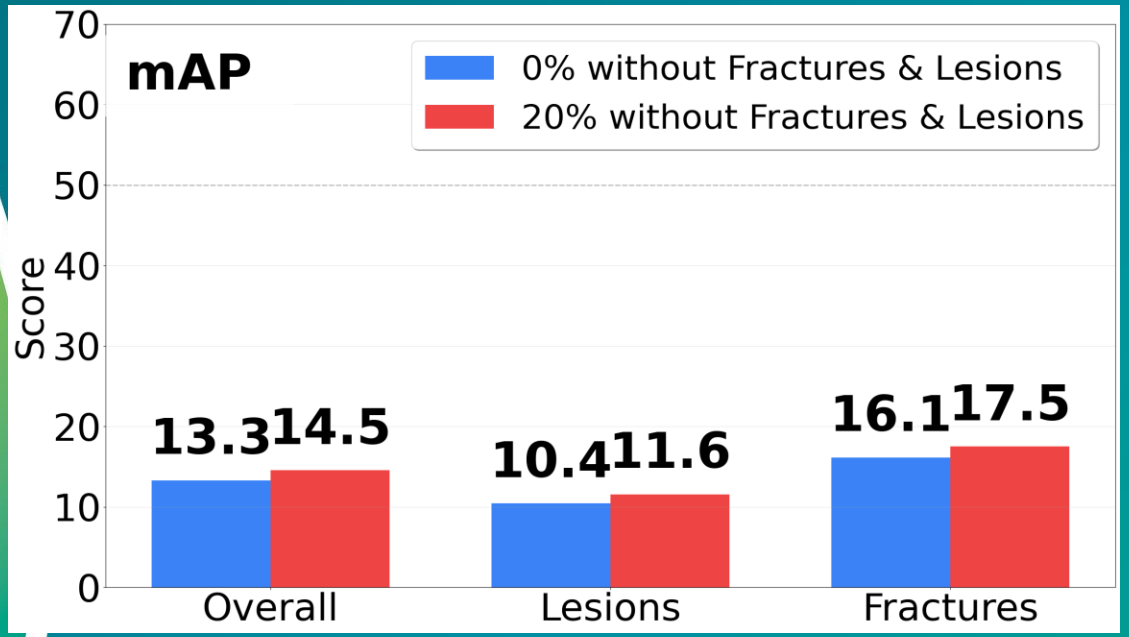
AP75: Average precision at IoU > 0.75

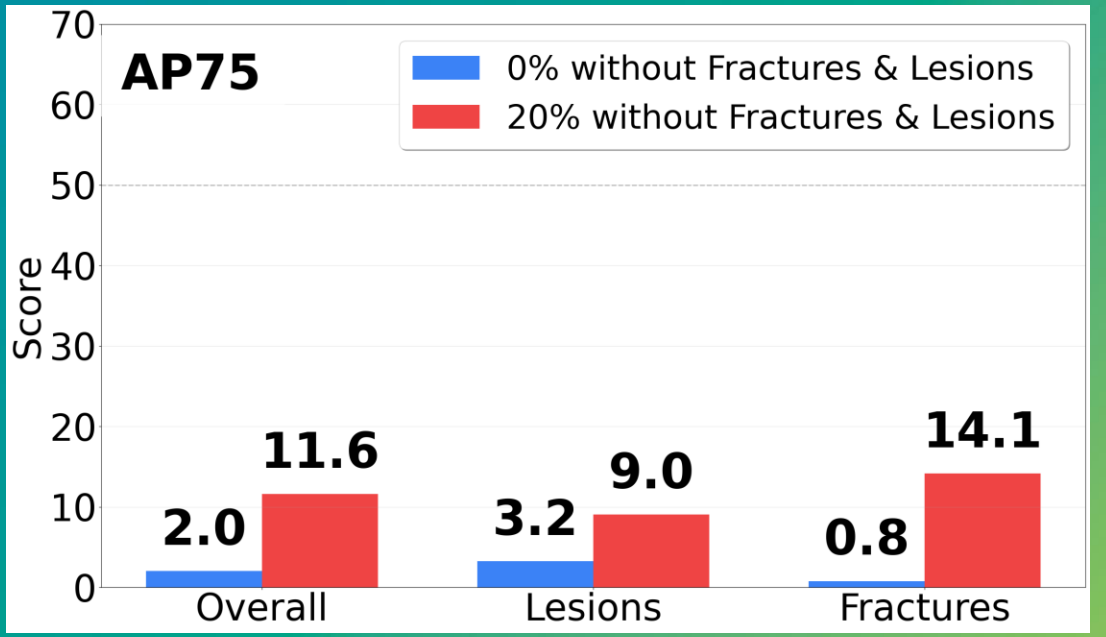
$$mAP = \frac{AP50 + AP55 + AP60 + AP65 + AP70 + AP75 + AP80 + AP85 + AP90 + AP95}{10}$$

Results: Held out Test

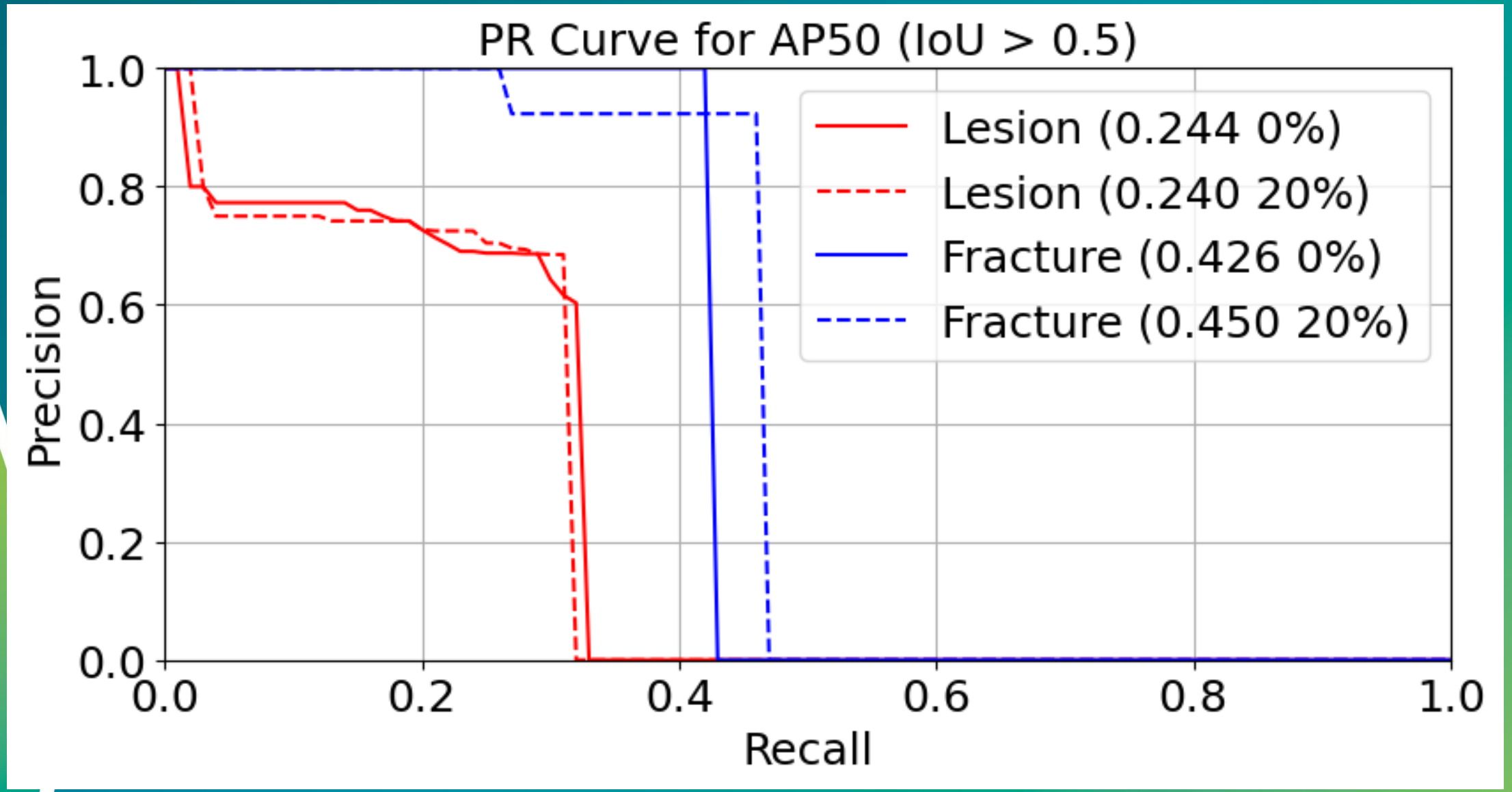
Summary:
1) Demonstrated reliable performance in detecting bone lesions and fractures.
2) 20% normal images > 0% normal





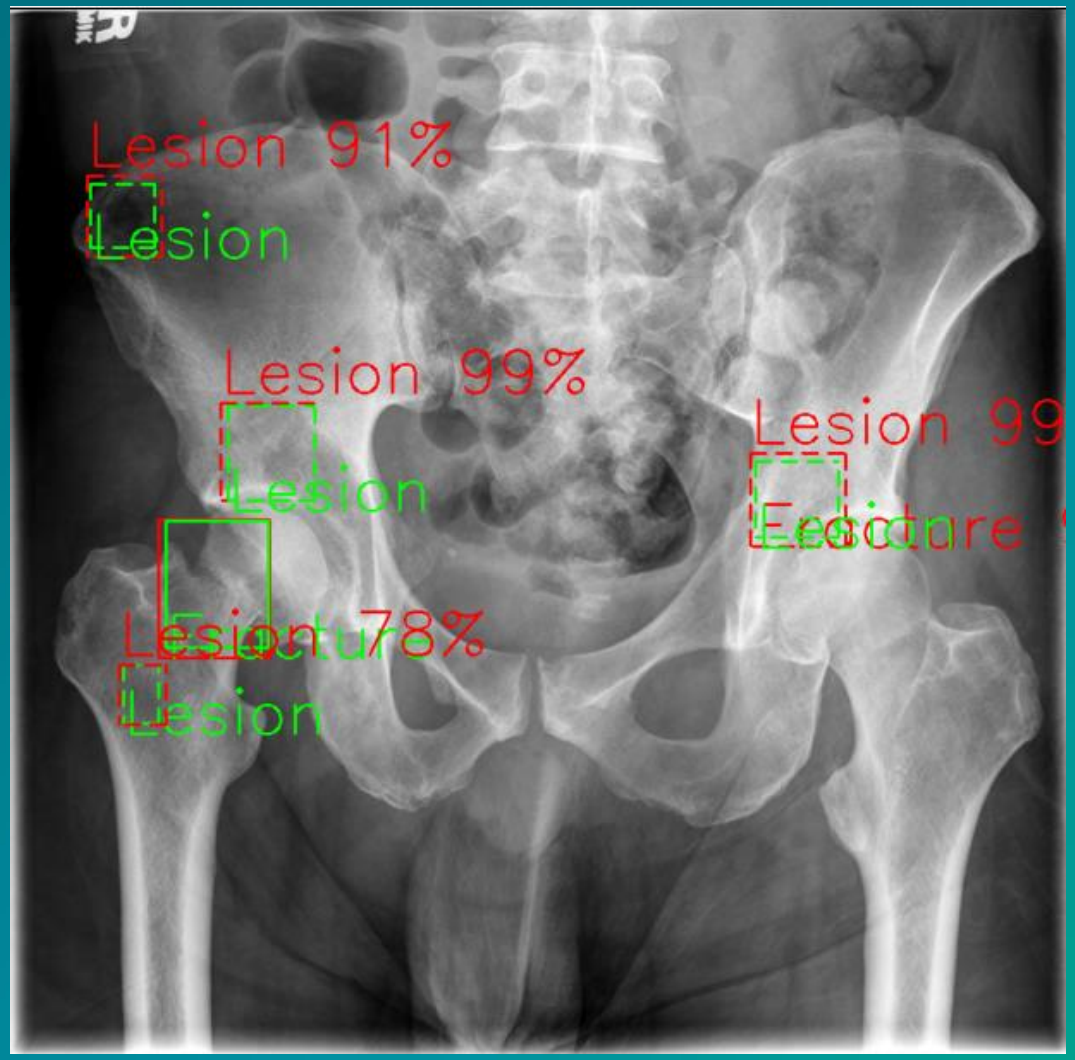


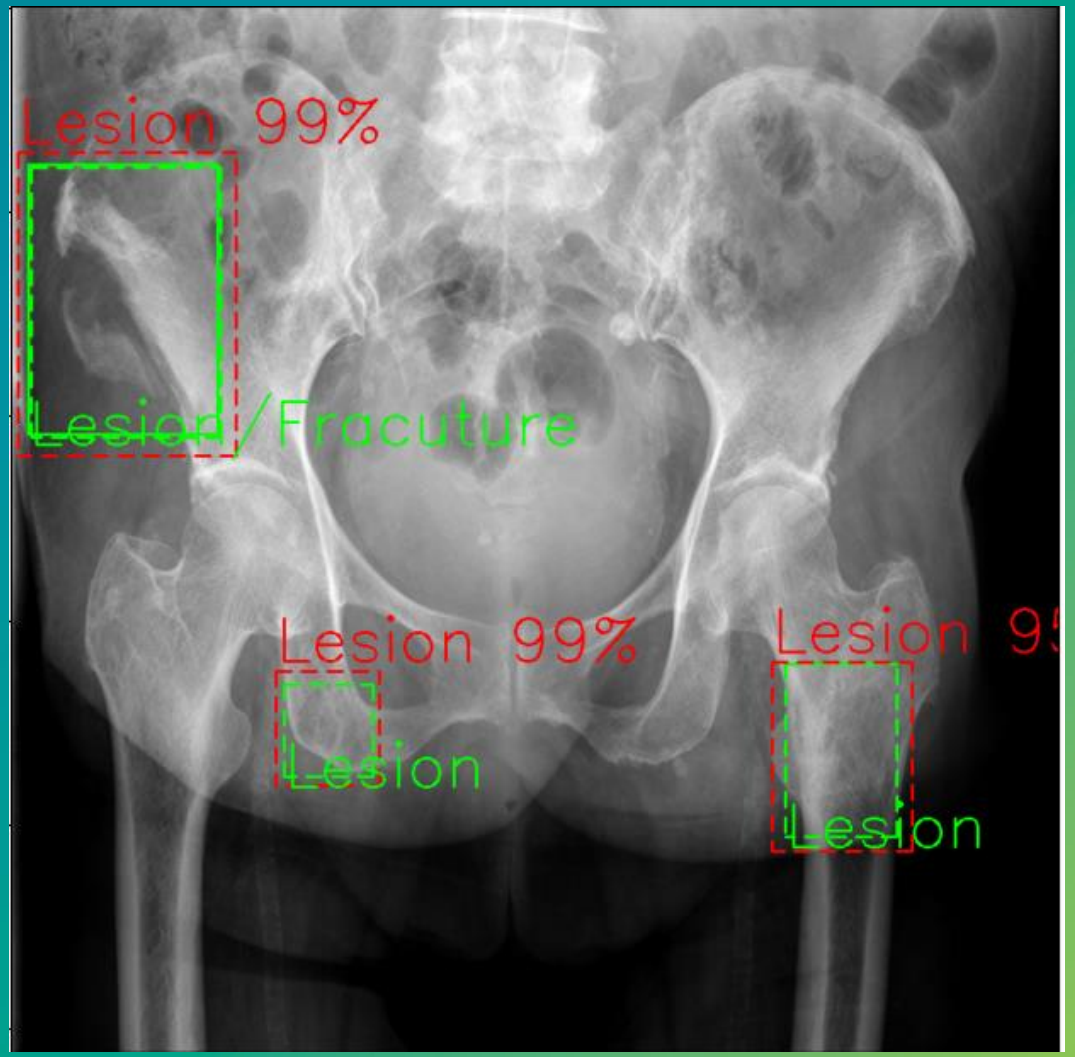
Results: PR curve



Results: Successful Predictions of Fracture / Bone Lesions

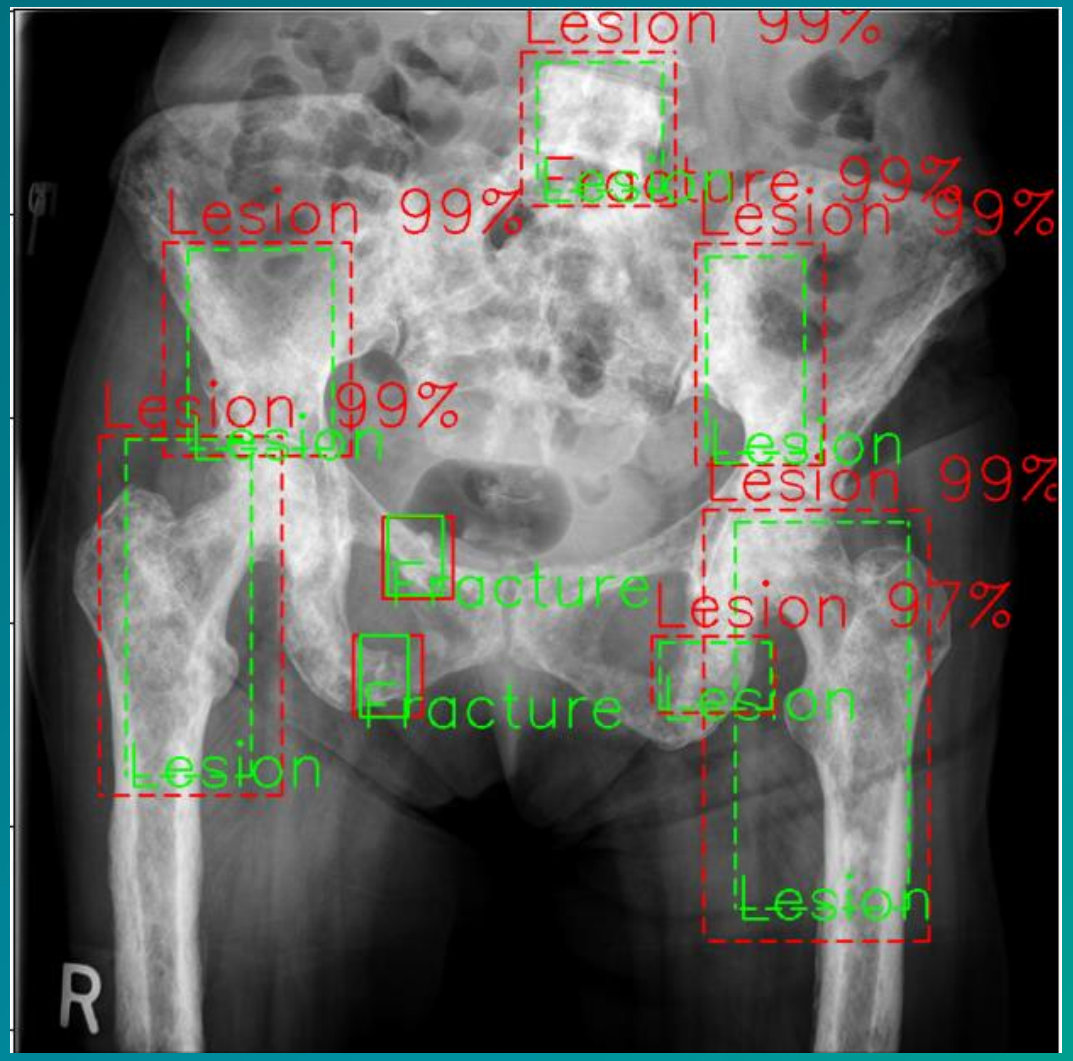
True Lesion
True Fracture
Predicted Lesion
Predicted Fracture

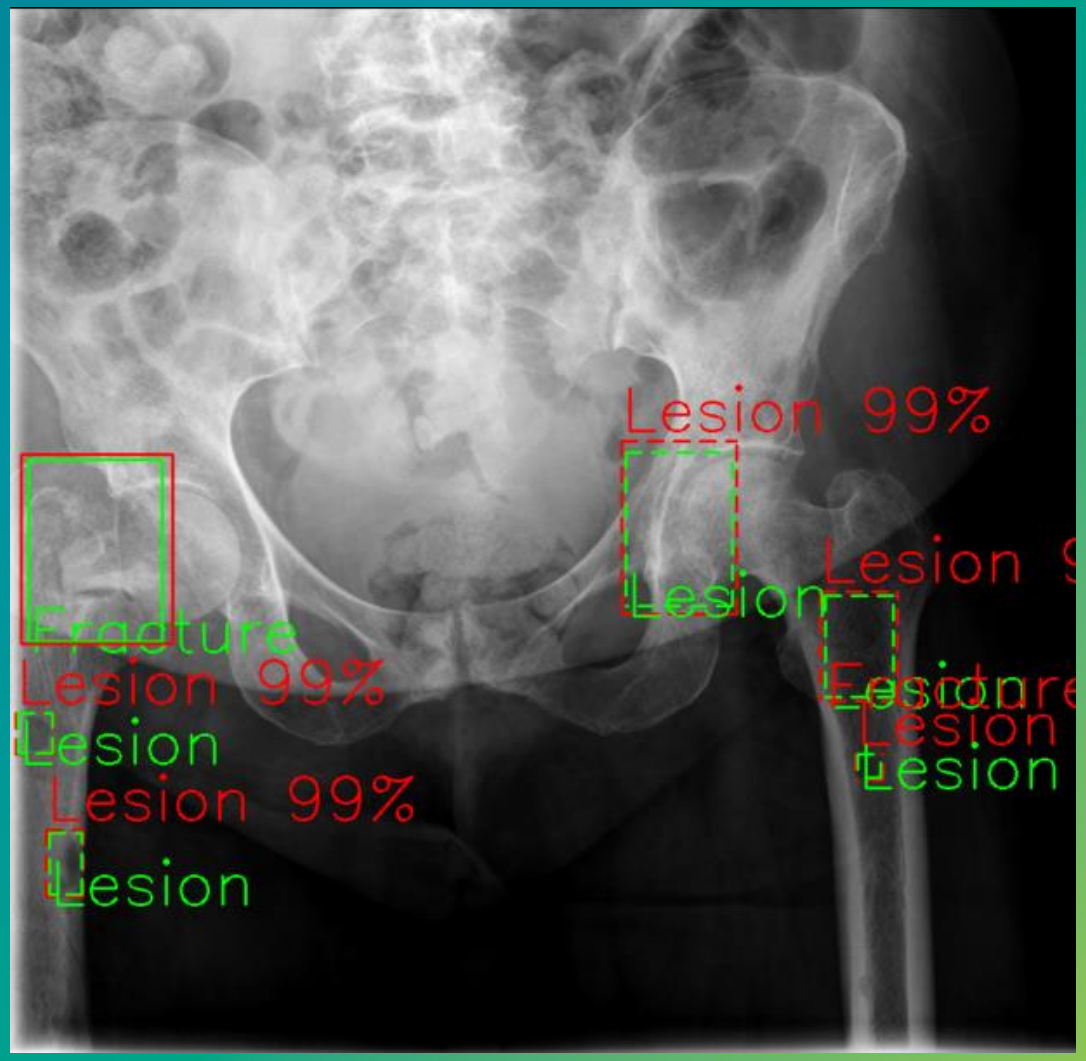




Results: Predictions on varying sizes / numbers

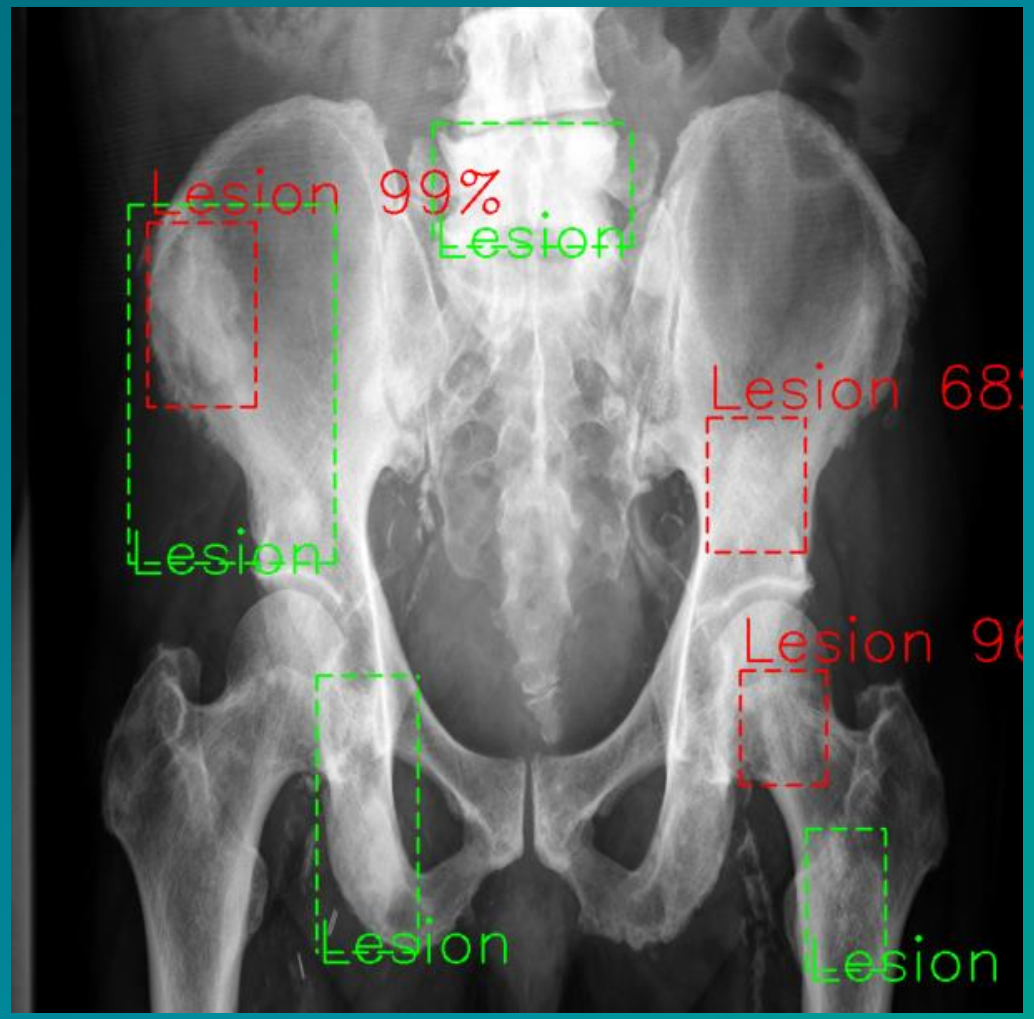
True Lesion
True Fracture
Predicted Lesion
Predicted Fracture

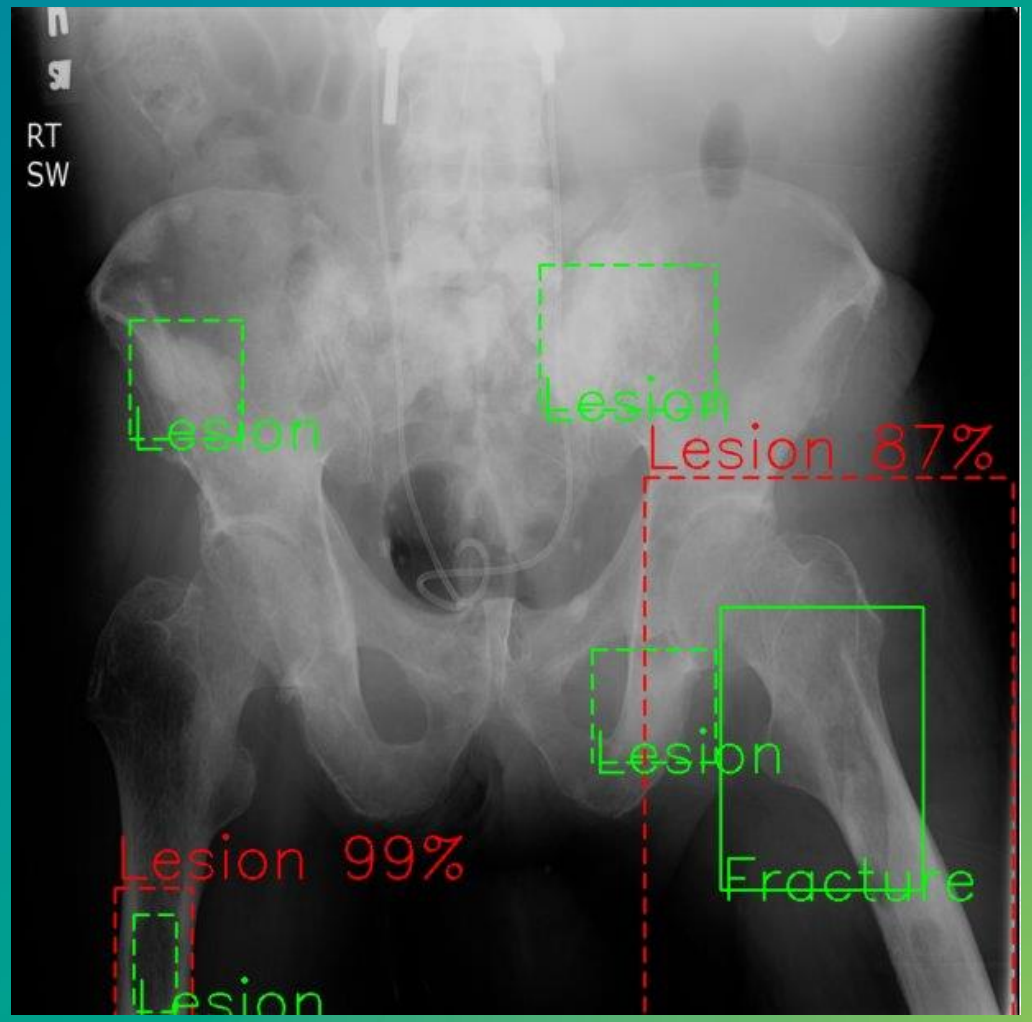




Results: Missed Predictions

True Lesion
True Fracture
Predicted Lesion
Predicted Fracture





Conclusions / Take away points

1)Successful Detection of Bone lesions and fractures

2)Fractures performance is higher than lesions

3)Adding 20% images without Fractures and Lesions ("Normal") improves overall performance

4)AI can streamline and standardize abnormality detection, potentially reducing interpretation time.

Nov. 30 – Dec. 4

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Thank You

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